Spark RDD作业



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◆学术论文的arXiv数据集: https://www.kaggle.com/Cornell-University/arxiv/version/62 (https://www.kaggle.com/Cornell-University/arxiv/version/62)

➢ 此数据集是原始 ArXiv 数据的镜像。由于完整数据集相当大（1.1TB 并且还在增长），因此此数据集仅提供格式的 json 元数据文件。此文件包含每篇论文的条目，包含：

- id: ArXiv ID (can be used to access the paper, see below)
- submitter: Who submitted the paper
- authors: Authors of the paper
- title: Title of the paper
- comments: Additional info, such as number of pages and figures
- journal-ref: Information about the journal the paper was published in
- doi: [https://www.doi.org] (https://www.doi.org%5D)(Digital Object Identifier)
- abstract: The abstract of the paper
- categories: Categories / tags in the ArXiv system
- versions: A version history

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```
import findspark

findspark.init()
```

- 1. spark 初始化

```
from pyspark import SparkConf,SparkContext

conf =

sc = SparkContext(conf=conf)

print(sc)
```

```
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(new
Level).

<SparkContext master=local[*] appName=rdd1>
```

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- 2. 读取数据

```
import json
rdd1 = sc.textFile( "…" )
rdd1 = rdd1.map(lambda x:json.loads(x)) # read json as json
rdd1.persist()
```

```
PythonRDD[16] at RDD at PythonRDD.scala:53
```

- 3. 查看核心数和分区数

```
rdd1.defaultParallelism
```

```
4
```

```
rdd1.getNumPartitions()
```

```
100
```

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- Question 1: Count elements

```
2011231
```

- Question 2: Get the first two records

```
[{'id': '0704.0001',
  'submitter': 'Pavel Nadolsky',
  'authors': "C. Bal\\'azs, E. L. Berger, P. M. Nadolsky, C.-P. Yuan",
  'title': 'Calculation of prompt diphoton production cross sections at Tevatron and\
n  LHC energies',
  'comments': '37 pages, 15 figures; published version',
  'journal-ref': 'Phys.Rev.D76:013009,2007',
  'doi': '10.1103/PhysRevD.76.013009',
  'report-no': 'ANL-HEP-PR-07-12',
```

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- Question 3: Get all attributes

```
['authors',
 'comments',
 'title',
 'id',
 'journal-ref',
 'versions',
 'submitter',
 'categories',
 'update_date',
 'authors_parsed',
 'report-no',
 'license',
 'abstract',
 'doi']
```

- Question 4: Get the name of the licenses

```
[None,
 'http://creativecommons.org/licenses/publicdomain/',
 'http://creativecommons.org/licenses/by-nc-nd/4.0/',
 'http://creativecommons.org/licenses/by-nc-sa/4.0/',
 'http://creativecommons.org/licenses/by-nc-sa/3.0/',
 'http://creativecommons.org/licenses/by/3.0/',
 'http://creativecommons.org/licenses/by/4.0/',
 'http://creativecommons.org/publicdomain/zero/1.0/',
 'http://arxiv.org/licenses/nonexclusive-distrib/1.0/',
 'http://creativecommons.org/licenses/by-sa/4.0/']
```

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- Question 5: Get the shortest and the longest titles

```
The shortest title : !-Graphs with Trivial Overlap are Context-Free
The longest title:  Weyl formula for the negative dissipative eigenvalues of Maxwell'
s
  equations
```

- Question 6: Find abbreviations with 5 or more letters in the abstract

-- 搜索:python regex w3

```
import re
def get_abbrivations(line):
        result = re.search(r"\(([A-Za-z][^_ /\\<>]{5,})\)", line)
        if result:
                return result.group(1)
                # return 1st match. group (0) will return all the matches
rdd1.filter(lambda x: get_abbrivations(x['abstract'])).count()# 192721
```

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```
rdd1.filter(lambda x: get_abbrivations(x["abstract"])).take(5)
```

```
[{'id': '0704.0055',
  'submitter': 'Per Hyldgaard',
  'authors': 'Eleni Ziambaras, Jesper Kleis, Elsebeth Schroder, and Per Hyldgaard',
  'title': 'Potassium intercalation in graphite: A van der Waals density-functional\n
study',
  'comments': '10 pages, 5 figures',
  'journal-ref': None,
  'doi': '10.1103/PhysRevB.76.155425',
```

- Question 7: Get the number of archive records per month

  --- --- ('update_date' attribute)



```
[(1, 134247),
 (2, 116948),
 (3, 126458),
 (4, 117126),
 (5, 296587),
 (6, 191746),
 (7, 122649),
 (8, 138469),
 (9, 138978),
 (10, 197755),
 (11, 297963),
 (12, 132305)]
```

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- Question 8: Get the average number of pages

```
the average of pages is:  17.85319004286046
```

```
sc.stop()
```